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Q: Research and present a timeline showing major milestones in ai history.

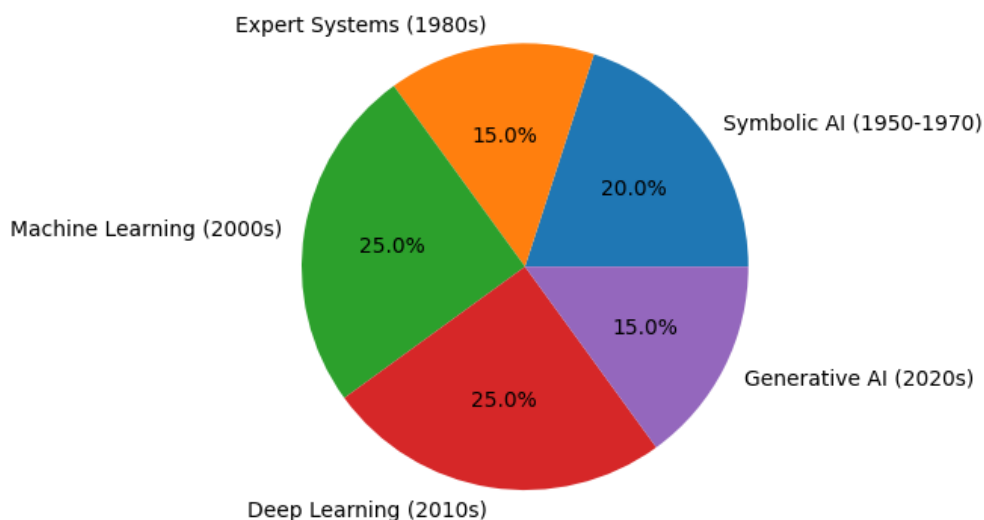
Introduction

Artificial Intelligence (AI) is a branch of computer science that focuses on creating machines and systems capable of performing tasks that normally require human intelligence. These tasks include learning from experience, reasoning, problem-solving, understanding natural language, recognizing images, and making decisions. The goal of AI is to develop intelligent systems that can simulate human thinking and behaviour.

The history of Artificial Intelligence dates back to the mid-20th century when scientists first began exploring whether machines could think like humans. Over the decades, AI has evolved through different stages, including symbolic reasoning, expert systems, machine learning, deep learning, and modern generative AI technologies. Each stage has contributed to significant advancements in computing and automation.

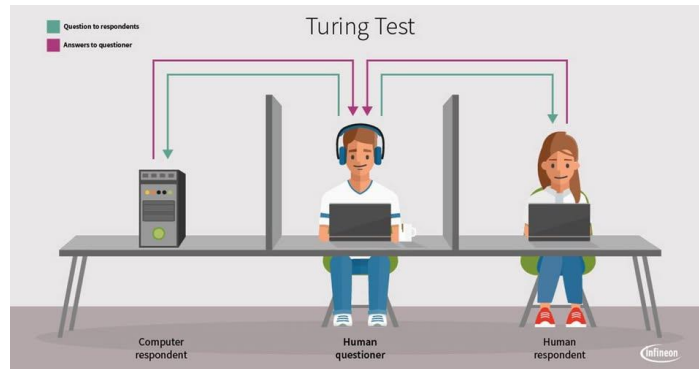
Today, AI plays an important role in everyday life. It is used in search engines, virtual assistants, healthcare systems, financial services, robotics, and self-driving vehicles. This timeline presents the major milestones in the history of Artificial Intelligence and highlights how it has developed from theoretical concepts into powerful real-world applications that continue to shape the future of technology.

Major Phases in the Evolution of Artificial Intelligence



1950 – Turing Test

1950



- Alan Turing proposed the Turing Test in his paper 'Computing Machinery and Intelligence'.
- The test evaluates whether a machine can imitate human responses convincingly.
- It became the philosophical foundation of Artificial Intelligence.
- Introduced the question: Can machines think?

1956 – Dartmouth Conference

1956



- The term 'Artificial Intelligence' was coined at Dartmouth College.
- John McCarthy organized the conference with other researchers.
- Marked the official beginning of AI as a research discipline.
- Focused on symbolic reasoning and problem-solving.

1960s–1970s – Early AI Programs

1960-1970

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Welcome to
          EEEEE LL   IIII ZZZZZZ AAAAA
          EE    LL   II   ZZ   AA  AA
          EEEEE LL   II   ZZZ  AAAAAA
          EE    LL   II   ZZ   AA  AA
          EEEEE LLLLL IIII ZZZZZ  AA  AA

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true, I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
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- ELIZA was an early chatbot simulating human conversation.
- SHRDLU demonstrated natural language understanding in blocks world.
- Research focused on rule-based and symbolic AI systems.
- Scientists believed human-level AI would soon be achieved.

1980s – Expert Systems

1980

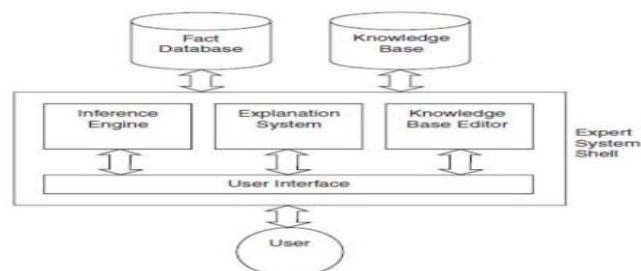


Fig Expert System Architecture

- Expert systems used knowledge bases and inference engines.
- MYCIN assisted doctors in diagnosing bacterial infections.
- Businesses adopted AI for decision-making.
- High costs and limitations led to the AI Winter

1997 – Deep Blue

1997

- IBM's Deep Blue defeated world chess champion Garry Kasparov.
- First time a computer defeated a reigning world champion.
- Demonstrated machine capability in structured strategic tasks.
- Boosted public interest in AI advancements.

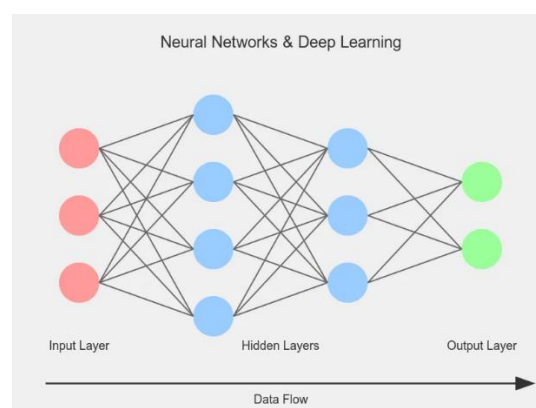
2000s – Rise of Machine Learning

2000

- Shift from rule-based systems to data-driven models.
- Machine Learning trained systems using large datasets.
- Growth due to big data and improved computing power.
- Used in search engines, fraud detection, and speech recognition.

2012 – Deep Learning Breakthrough

2012

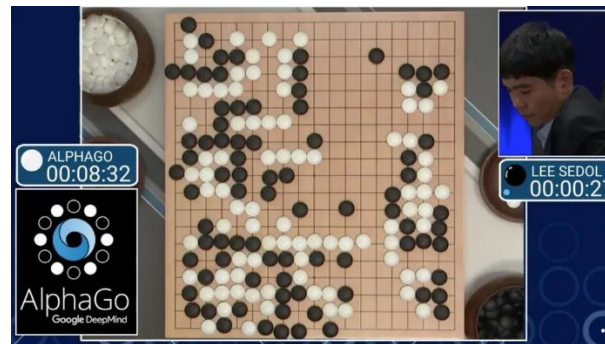


- AlexNet won the ImageNet competition using deep neural networks.
- Significantly reduced image classification error rates.
- Triggered rapid growth in deep learning research.

- Led to improvements in speech and vision recognition.

2016 – AlphaGo

2016



- Developed by DeepMind using reinforcement learning.
- Defeated world Go champion Lee Sedol.
- Proved AI can master highly complex games.
- Combined neural networks with search algorithms.

2020 – GPT-3

2020



- GPT-3 introduced advanced natural language generation.
- Trained on massive datasets with billions of parameters.
- Enabled applications like chatbots and content generation.

- Marked the rise of large language models (LLMs).

2022–Present – Generative AI Era

2022



- AI systems can generate text, images, and code.
- Widely used in healthcare, education, finance, and robotics.
- Multimodal AI combines text, vision, and audio understanding.
- Raises ethical concerns regarding bias and data privacy.

Conclusion

The history of Artificial Intelligence shows a journey from theoretical ideas in 1950 to advanced generative systems today. Over the decades, AI evolved through expert systems, machine learning, deep learning, and large language models. Each milestone contributed to making AI more powerful and practical. Today, AI transforms industries and daily life, and its future development requires responsible innovation and ethical regulation.